IMPERIAL

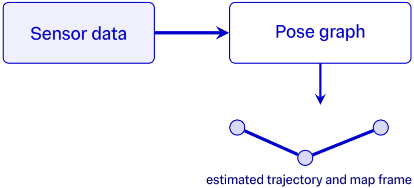
Sonar-Based Underwater SLAM with 3D Gaussian Splatting Mapping

Saxon Shang supervised by Dr. Sen Wang
June 24th 2026

SLAM and 3D Gaussian Splatting Mapping

Two complementary building blocks

Simultaneous Localisation and Mapping



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- Fuses motion and observations to reduce accumulated drift.

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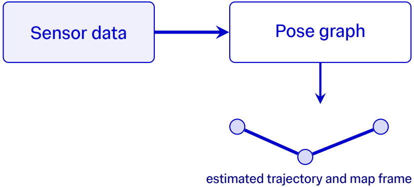

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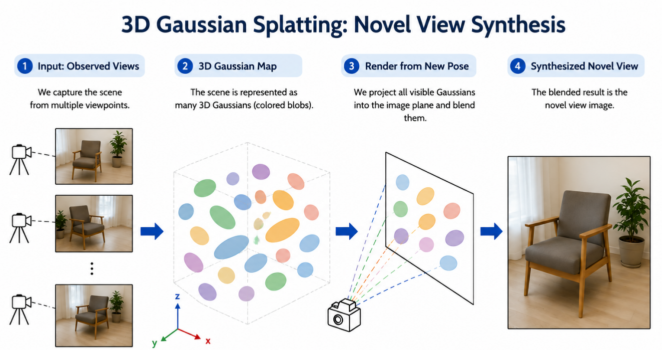
3D Gaussian Splatting Mapping



Why Underwater Sonar SLAM Is Hard

Can a renderable sonar map provide online correction factors?

1. Problem setting

Underwater robots cannot use GNSS. IMU/DVL odometry therefore drifts over time.

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6. Project idea

Use render consistency between live sonar and 3D Gaussian Splatting prediction as a SLAM constraint.

Research Gap: Sonar SLAM + 3DGS

Three active research lines, one missing connection

Underwater SLAM

- Fuses sonar, camera, IMU, and DVL.
- Uses feature, image, or sensor factors.

Li et al. 2018; Cardaillac and Ludvigsen 2023; Xu et al. 2025.

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- Focuses on reconstruction and rendering, not SLAM.

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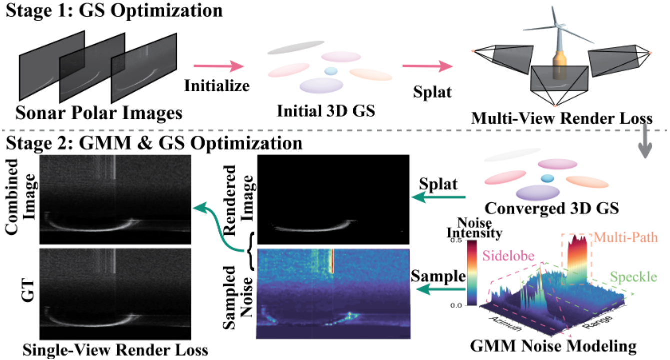
Missing integration in the reviewed literature

Live sonar $\rightarrow$ sonar 3DGS candidate render $\rightarrow$ gated Pose3 factor $\rightarrow$ iSAM2.

NAS-GS: Noise-Aware Sonar Gaussian Splatting

A renderable sonar measurement model reference



NAS-GS training pipeline, adapted from Xu et al.

What NAS-GS does

- Learns a 3D Gaussian sonar scene from polar sonar images.
- Optimises rendered structure together with acoustic noise appearance.
- Produces a differentiable sonar renderer for new viewpoints.

How I use it

- Freeze the trained map during replay.
- Render candidate Pose3 hypotheses online.
- Insert a factor only when live sonar and render agree.

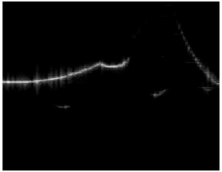
SLAM Pose graph optimization with Pose3 constraints and iSAM2 backend

NAS-GS predicts observations; iSAM2 consumes accepted constraints

Online loop: render from the map, compare with the live frame, then add a graph factor only when the evidence is strong.

Renderable sonar map

NAS-GS predicts what the sonar should observe from a proposed Pose3 camera pose.



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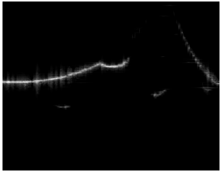
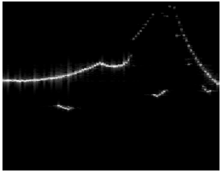
Online evidence

A live sonar frame is compared against rendered candidates around the current drifted pose.





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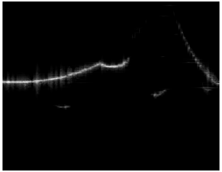
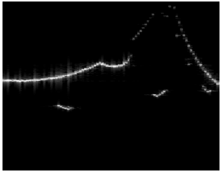
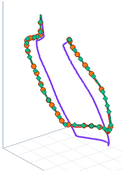
Graph correction

If render similarity and gates pass, the recovered pose is inserted as a Pose3 factor.







Online-render Registration Backbone

Same online loop for same-route correction and revisit correction

Live replay sonar + Pose3 → Candidate poses → NAS-GS render → Score and gate → Pose3 factor → iSAM2 update

Local single

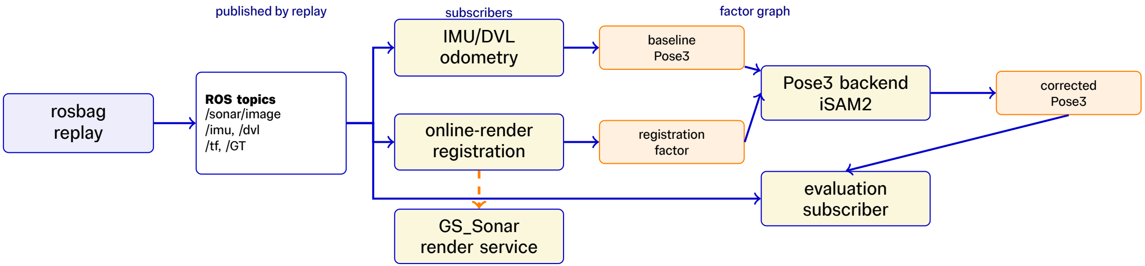
Compact x, y, z, ψ search around the live Pose3 baseline. Main same-route policy for Pass A and off-shore replay.

Route-progress revisit

Sequence render search guided by route progress and drift-prior checks. Used for revisit cases such as Pass B.

Gating is separate from the mode. Relaxed gating accepts useful factors earlier, while score and structure checks still prevent weak render matches from entering the graph.

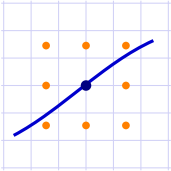
Algorithm Step 1: Live Replay System



The replay is deliberately online. Nodes subscribe to ROS topics in time order, the registration node queries NAS-GS only at candidate poses, and iSAM2 receives only gated Pose3 factors.

Algorithm Step 2: Candidate Pose And NAS-GS Render

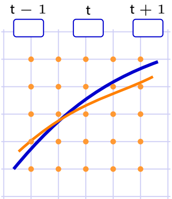
Local single-frame search



single live frame
around baseline

Compact x, y, z, ψ grid for same-route correction.

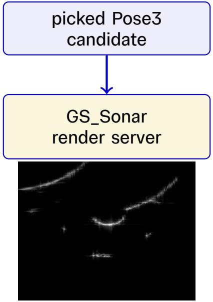
Route-progress sequence search



sequence evidence
around route prior

The orange line is a smooth candidate correction sequence. Wider candidates are allowed, but neighbouring frames must agree.

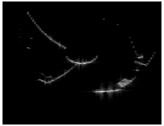
Render query



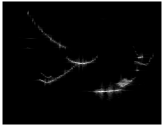
The output is a rendered sonar image for the proposed robot pose.

Algorithm Step 3: Score And Gate

Accepted pair



live sonar



rendered sonar

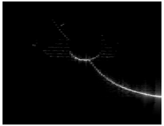
Frame 24 selects rendered frame 24 and passes the score gate.

score gate	s $= 0.926 \geq 0.08$, pass
pose gate	shift $= 0.048$ m, pass
replay evidence	published as a Pose3 registration factor

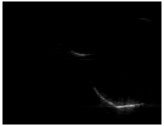
Score calculation

Intensity normalisation, NCC-style similarity, residual penalty, and bright-return overlap are combined into one render score. Relaxed threshold $s_{\mathrm{min}} = 0.08$.

Rejected pair



live sonar



rendered sonar

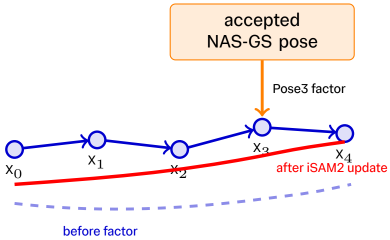
Frame 228 selects rendered frame 216 but falls just below threshold.

score gate	s $= 0.0799 < 0.08$, fail
edge check	best candidate on search edge
replay evidence	one of 6 score-below-threshold events

Gates before insertion

Render-score, structure-support, pose-jump, and temporal-consistency gates must pass before publishing a Pose3 factor to iSAM2.

Algorithm Step 4: Pose3 Factor And iSAM2 Update



Before

The graph contains IMU/DVL odometry factors, so drift accumulates along the chain.

Insertion

The accepted candidate pose and covariance are published as a Pose3 registration factor.

After

iSAM2 optimises the graph, pulling nearby states toward the render-consistent pose.

Algorithm Step 5: Route-progress Revisit Priors

Pass B window → route progress selects map window → drift prior checks envelope → sequence render search → score and gate → Pose3 factor

Route prior

$$s_t = \frac{t}{N_B - 1}, \qquad i_t = \mathrm{round}(s_t(N_A - 1))$$

Variables: t is the current revisit frame, N_A, N_B are the map and revisit route lengths, s_t is normalised progress, and i_t is the selected Pass A map index.

- This avoids nearest-pose lookup when route direction changes.
- It gives a map window, not the final factor pose.

Drift prior

$$\bar{e}(d) = \mathrm{PCHIP}\Big(d_j, \big|\mathrm{smooth}_{21}(e_j)\big|\Big), \qquad e_j = \mathrm{GT}_{A,j} - \mathrm{base}_{A,j}$$

$$r = \max(0, |\Delta T_B| - \bar{e}(d_B))$$

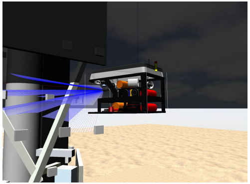
Variables: d_j is accumulated Pass A baseline XY distance; $e_j = [e_x, e_y, e_z, e_\psi]$ is its aligned GT correction; $\bar{e}(d)$ is the expected absolute correction; ΔT_B is a Pass B candidate minus its live drifted pose.

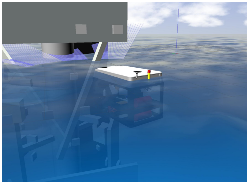
- Align Pass A baseline and GT, then smooth over 21 samples (about 10.7 s or 1.5 m): enough to suppress local jitter while retaining accumulated drift.
- Fit a component-wise PCHIP curve against travelled XY distance and query it at the current Pass B distance.
- Manual transfer slack on the excess, not the total correction: 3.0 m XY, 2.5 m z, and 0.70 rad yaw.
- Absolute magnitudes transfer across route direction; NAS-GS score still selects the final pose.

Simulation Platform: DAVE In Docker
Repeatable sonar, IMU, DVL, and ground-truth recording

Docker setting and robot model

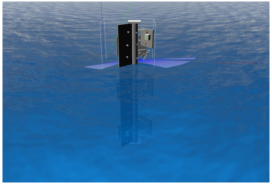
- CUDA-enabled Docker container.
- DAVE with ROS Noetic and Gazebo.
- BlueROV2 vehicle model.
- Sensors recorded: sonar, IMU, DVL, odometry, and pose ground truth.





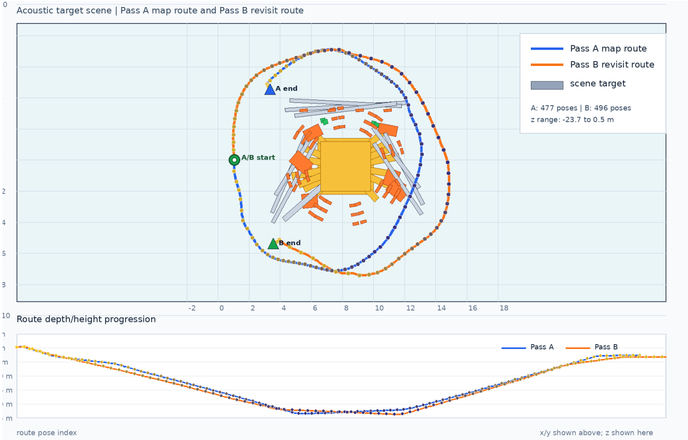
Custom scene description

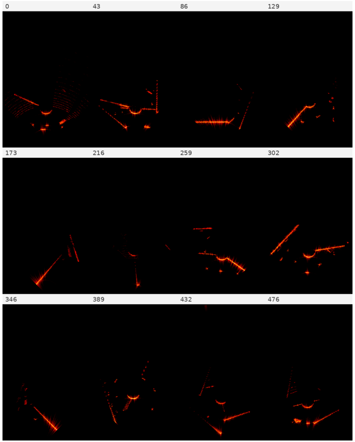
- A fixed underwater target structure creates repeatable sonar returns.
- Pass A, Pass A revisit, and Pass B use controlled waypoint routes.
- Sonar field of view, range, beam pattern, and noise are explicitly set.



Simulation Path Recording

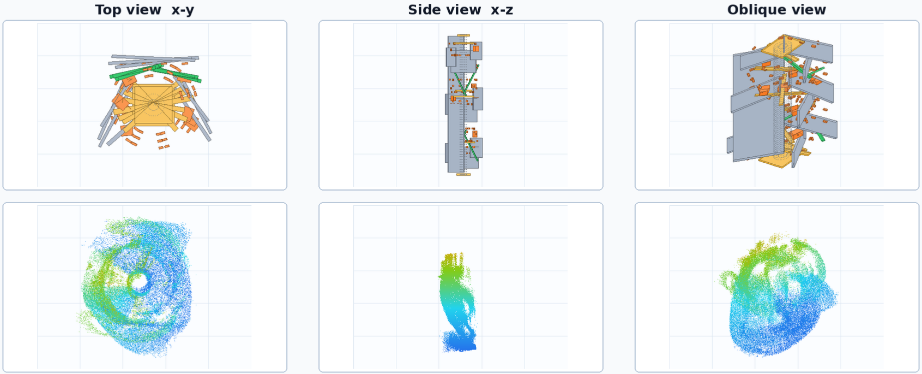
Pass A trains the map; Pass B reuses the start point but follows a hard revisit path





From Simulated Geometry To NAS-GS Map
The learned map

The two rows use the same top, side, and oblique projections. They differ because the DAVE row shows the full simulated target geometry, while the NAS-GS row shows Gaussian centres learned only from sonar-visible acoustic returns.



Top row: DAVE target geometry from SDF. Bottom row: NAS-GS map centres from point_cloud.ply. Same projection and bounds per column.

Experimental Design

Recorded data, map training, and replay policy

Case	Images	Sonar setting	NAS-GS map	Replay policy
Pass A	477	90°, 20 m	Pass A 900k/i1000, PSNR 35.77 dB	Relaxed local single
Pass A revisit	431	90°, 20 m	Pass A 900k/i1000, PSNR 35.77 dB	Relaxed local; route-progress check
Pass B revisit	496	90°, 20 m	Pass A 900k/i1000, PSNR 35.77 dB	Relaxed route-progress sequence
Offshore	1139	120°, 12 m	Offshore 750k/i800, BP PSNR 33.68 dB	Relaxed local single

Baseline setting

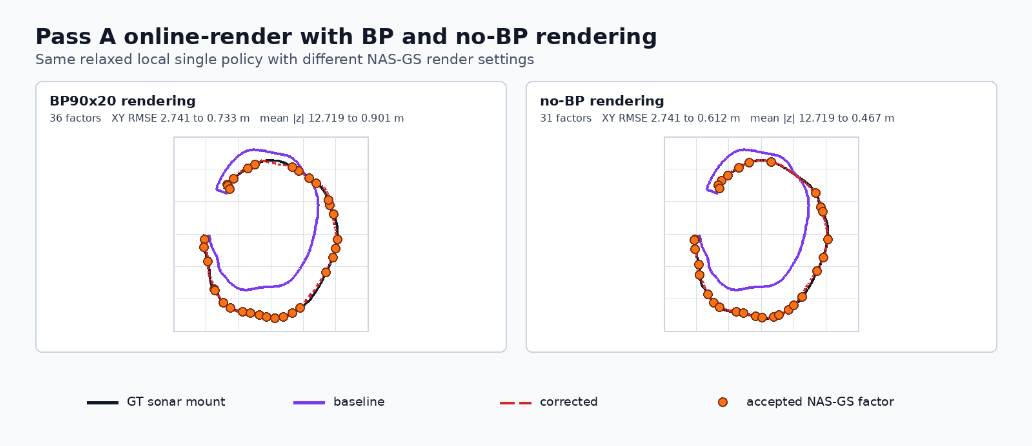
IMU/DVL Pose3 replay with NAS-GS factors disabled. Noise is injected as DVL scale 1.10, body bias $(0.030, -0.025, 0.0)$ m/s, yaw drift 8.0×10^{-4} rad/s, and 2.5 s DVL dropout every 35 s after 45 s.

Relaxed gating

Minimum render score 0.08, maximum accepted correction 5 m in x, y and 8 m in z, followed by render-validity and factor-consistency checks.

Pass A Policy Selection

Rendered BP library with runtime no-BP was the most reliable local policy

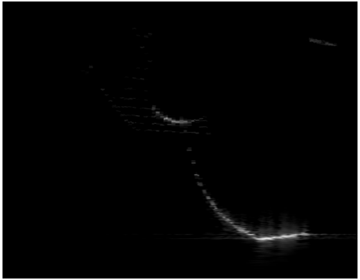


Pass A Sonar Evidence

One recorded frame and the corresponding BP/no-BP rendered libraries

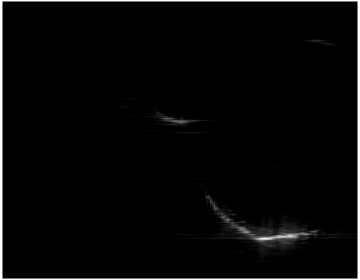
Recorded Pass A

Live sonar frame from the DAVE route.



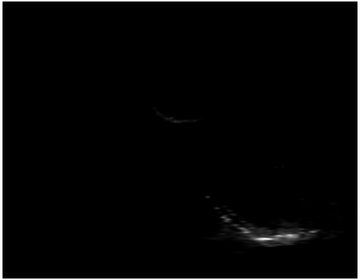
Rendered BP library

Reference image from the trained NAS-GS map.

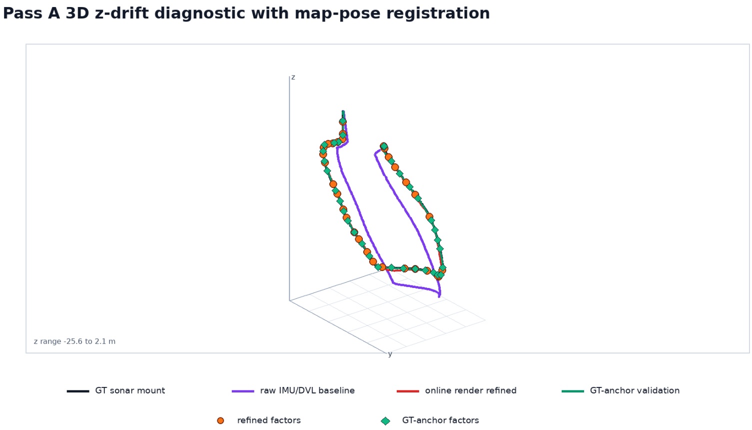


Rendered no-BP library

Alternative rendered appearance for policy testing.



Pass A Result



Metrics: 30 factors. XY RMSE $2.741 \to 0.570$ m; XYZ RMSE $15.078 \to 0.937$ m; mean |z| $12.719 \to 0.569$ m.

Final error: XY 0.305 m; |z| $= 0.069$ m.

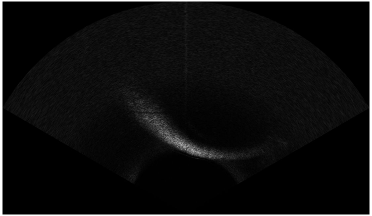
Offshore Dataset: Real Sonar Validation

151216 wind-turbine inspection sequence

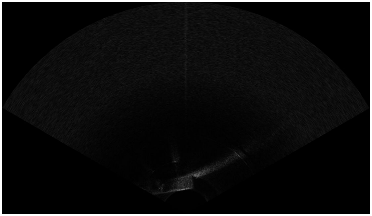
- Real 151216 sequence recorded around an offshore wind-turbine foundation with a Tritech Gemini 1200ik imaging sonar.
- More than 20 minutes and **1,139 frames** from the surface towards the seabed, using a 120° field of view and 12 m range.
- Sidelobe, speckle, multipath, and strong vertical motion make this a demanding real-acoustic transfer case.

Reference and validation role

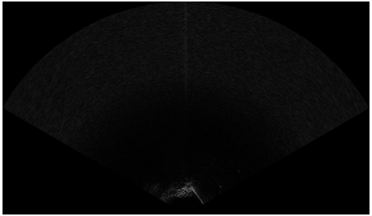
Aqua-SLAM 6-DoF poses supervise NAS-GS and provide the replay reference. This is not independent survey ground truth.



Frame 216: strong curved return



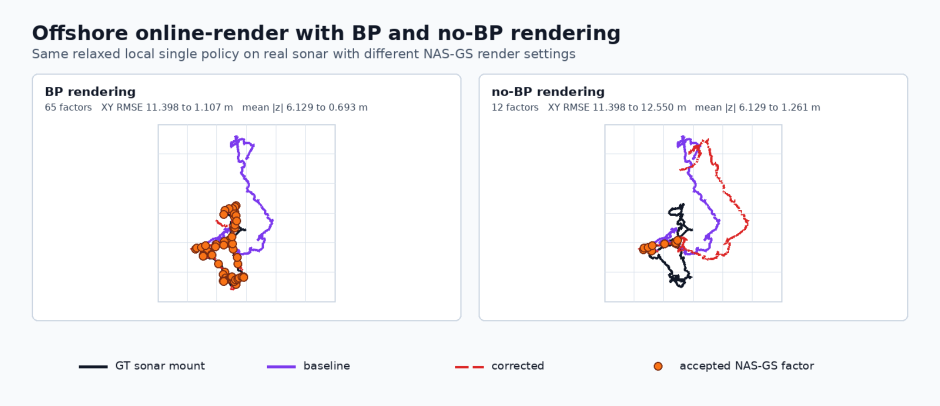
Frame 517: partial return



Frame 864: sparse near-range return

BP versus no-BP on Offshore Data

The better rendering mode depends on the recorded sonar appearance



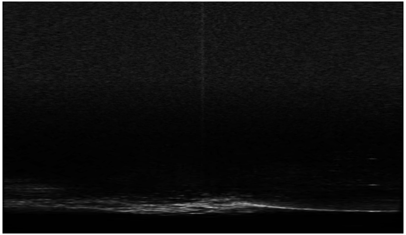
Offshore data contains stronger real-image artefacts, so BP and no-BP rendering are less interchangeable than in the DAVE Pass A replay.

Offshore Sonar Evidence

The real replay has stronger appearance mismatch than the DAVE route

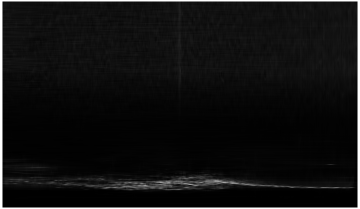
Recorded offshore

Live sonar from the replay.



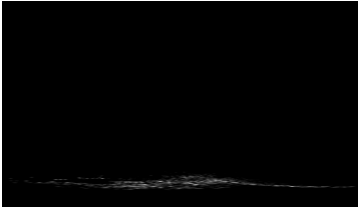
Rendered BP library

NAS-GS render with the offshore BP library.



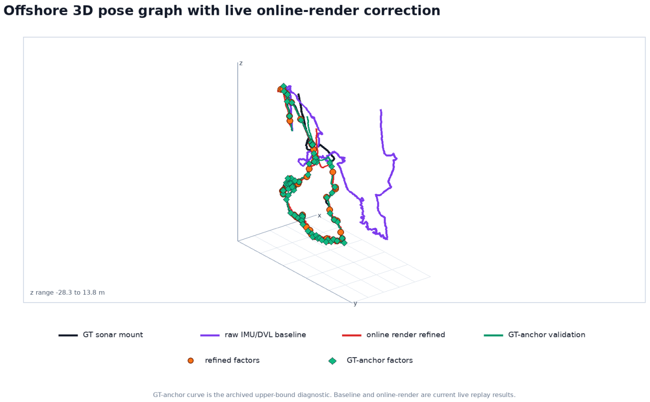
Rendered no-BP library

No-BP render used for appearance comparison.



Offshore Validation

The method transfers, but the real-scene case exposes z and appearance limits

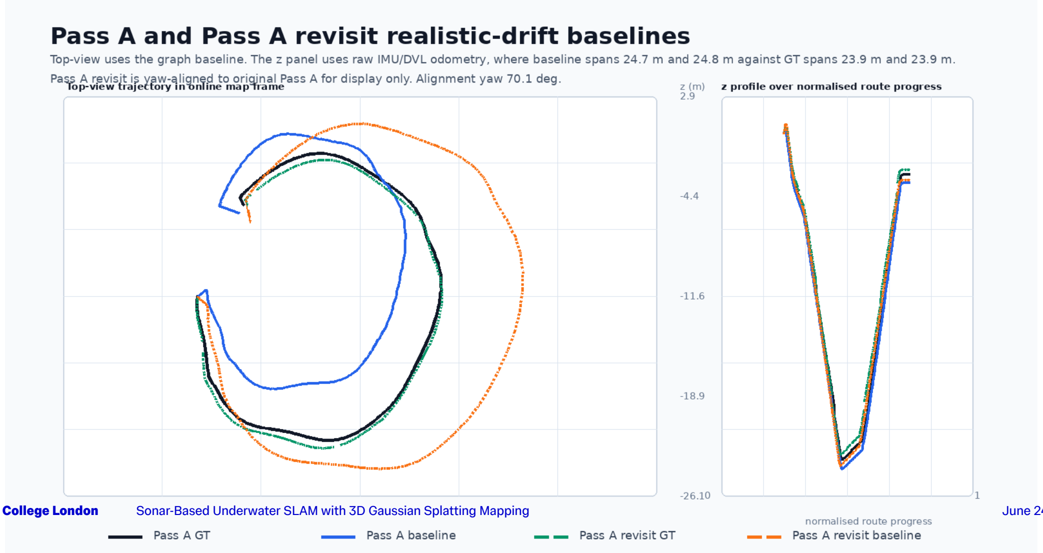


Metrics: 50 factors. XY RMSE $11.043 \to 1.407$ m; XYZ RMSE $12.795 \to 2.126$ m; mean |z| $6.074 \to 1.027$ m.

Final error: XY 9.843 m; |z| $= 3.601$ m.

Revisit Baselines

Pass A revisit is same-route; Pass B is harder because the route direction changes



Revisit Result

The revisit policy improves the hard case conservatively

Route-progress and same-route revisit replay results

Local single is strong for the same-route Pass A revisit. Route-progress gives guarded partial correction for the harder revisit cases.

Limitations and Future Work

The next steps follow directly from the observed failure modes

Current limitations

The remaining weaknesses define the operating boundary.

- Reverse-route revisit remains sensitive to route-progress endpoints and large drift.
- Wide candidate searches require stronger temporal evidence before large corrections are safe.
- Offshore evaluation uses Aqua-SLAM as the reference rather than independent survey ground truth.

Key future work

Each direction targets one demonstrated limitation.

- Develop sequence-level optimisation for robust large-drift recovery.
- Stabilise route-progress estimation at reverse-route boundaries.
- Retrain NAS-GS using corrected trajectories and validate against an independent reference.
- Progress from live replay to onboard timing, calibration, and safety trials.

Main Contributions

- **Integrated simulation framework** DAVE data collection, NAS-GS training, and a live ROS Pose3 SLAM system.
- **Real-data-validated registration policies** Controlled IMU/DVL baselines and gated online-render factors for pose-graph correction.
- **Foundation for sonar-based 3DGS SLAM** A working connection from renderable sonar maps to incremental iSAM2 constraints.

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Thank you

Questions?